



Towards a circular plastics economy: synchronising material design, hybrid processing, digital logistics, and adaptive policy

Maneesha P. Ginige · Andrew C. Warden ·
Anna H. Kaksonen

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Abstract The escalating environmental cost of global plastic production is driven by a fundamental misalignment: the complexity of modern polymer chemistry has outpaced the capability of linear waste management infrastructure. Addressing this crisis requires moving beyond fragmented mechanical and thermal solutions to a fully integrated industrial framework that synchronises material innovation with biological discovery. This review articulates a strategic roadmap to transition from a linear disposal model to a robust bio-industrial circular economy, with a predominant focus on the deployment of emerging bio-catalytic and bio-hybrid processing systems. We distinguish between the dual goals of resource recovery (circularity) and

safe mineralisation (environmental resilience). Four interdependent pillars essential for this transition are identified: (1) Material design, where “design for degradation” is embedded at the molecular level; (2) Bio-hybrid processing, which supersedes single-mode recycling by synergising biological selectivity with physicochemical throughput (e.g., chemo-biological and photochemical-biological coupling) to handle mixed waste streams; (3) Digital logistics, utilising the “Internet of materials” to enable high-resolution sorting and decentralised processing; and (4) Adaptive policy, where standards are co-developed to verify system compatibility and increased stakeholder engagement. A “paradigm shift” is necessary to align these domains. Only by integrating the material, the process, the data, and the policy can plastic waste be transformed from an environmental liability into a predictable, high-value bio-industrial resource.

M. P. Ginige (✉) · A. H. Kaksonen
CSIRO Environment, 7 Conlon Street, Waterford,
WA 6152, Australia
e-mail: maneesha.ginige@csiro.au

A. C. Warden
CSIRO Environment, Clunies Ross Street, Canberra,
ACT 2601, Australia

A. H. Kaksonen
Western Australian School of Mines: Minerals,
Energy and Chemical Engineering, Faculty of Science
and Engineering, Curtin University, Bentley, WA 6102,
Australia

A. H. Kaksonen
School of Engineering, University of Western Australia, 35
Stirling Highway, Nedlands, WA 6009, Australia

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1 Introduction

1.1 The global plastic crisis: production, pollution, and the logistics bottleneck

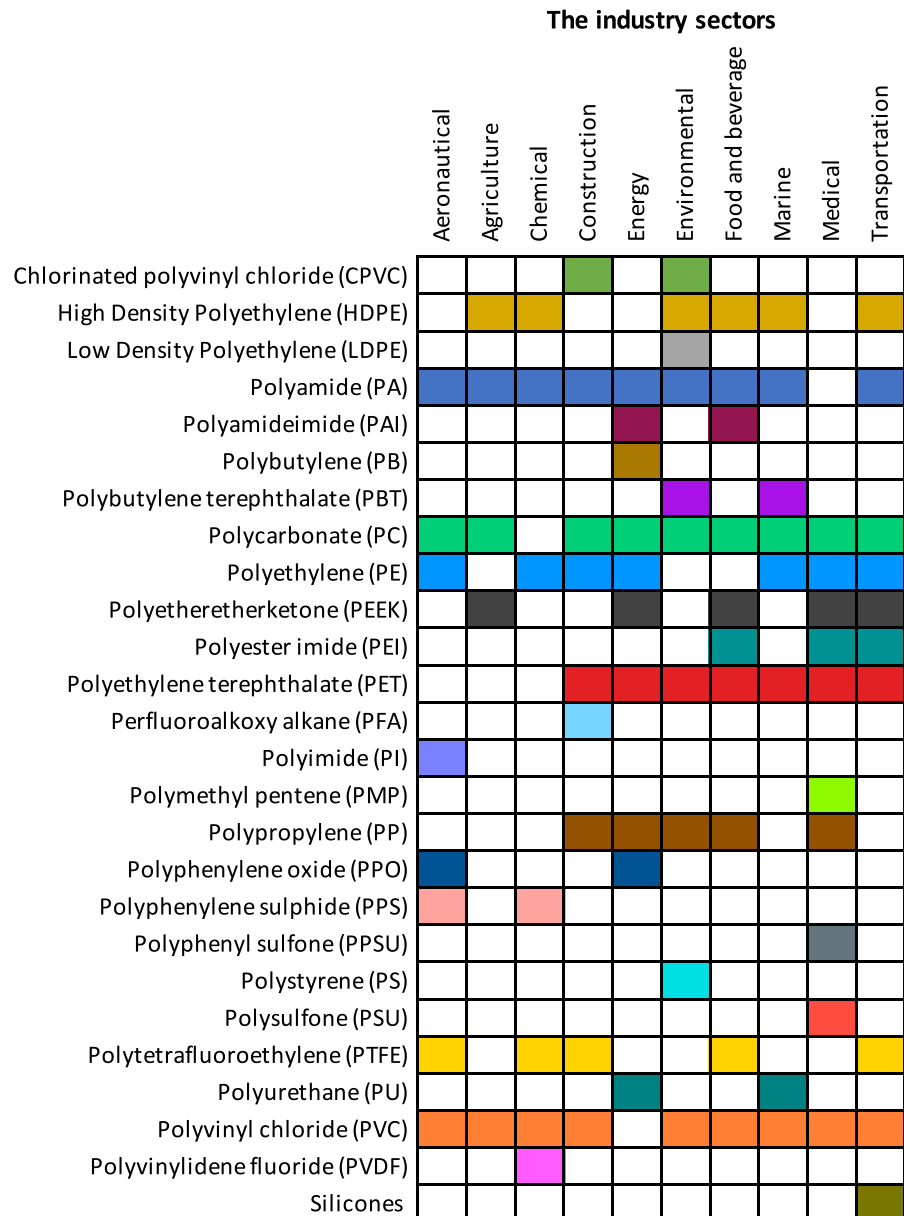
The proliferation of synthetic plastics, valued for their durability, longevity, light weight, and affordability, has rendered them indispensable across a wide range

of human and natural systems worldwide (Chbib et al. 2019; Jambeck and Walker-Franklin 2023). This has created a significant threat to the environment with plastic contamination reaching even the most remote parts of the world, including isolated polar regions, uninhabited islands, and deep ocean trenches (Barnes et al. 2009; Borrelle et al. 2017). Global plastic production has experienced a dramatic increase, escalating from an annual 1.5 million metric tonnes (Mt) in 1950 to an annual 350–367 Mt by 2017–2020

(Amirmatova and Aliyev 2025; Hoang 2022). Driven by demand across diverse sectors from packaging to construction (Fig. 1) production is projected to exceed 1,200–1,600 Mt annually by 2050 (Château et al. 2020; Reis et al. 2017) if current trends persist.

This exponential growth has precipitated a multifaceted crisis involving resource depletion, escalating greenhouse gas emissions, and severe environmental contamination. The industry currently utilises a significant portion of global oil consuming 4% as

Fig. 1 Major plastics used by various industry sectors (Adapted from Kumar et al. (2020))



feedstock and an additional 3–4% for energy (Kabeyi and Olanrewaju 2023) with demand projected to encompass 20% of the world's total oil by 2050 (United Nations Environment Programme (UNEP) 2018). This reliance drives a corresponding surge in greenhouse gas (GHG) emissions, which are expected to rise from 1,781 Mt CO₂ equivalent in 2015 to 6,500 Mt by 2050 without intervention (Zheng and Suh 2019). The scale of this challenge is underscored by the fact that, of approximately 6,300 Mt of plastic waste generated globally up to 2015, only 9% was recycled and 12% incinerated, leaving the remaining 79% to accumulate in landfills or the natural environment (Geyer et al. 2017). If current trends persist, roughly 12,000 Mt of plastic will be discharged into the environment by 2050 (Adekomaya and Majozi 2020), further exacerbating the proliferation of microplastics, which already constitute approximately 13% of environmental plastic waste that is estimated to enter the oceans at a rate of over 20,000 tons per day (Zhang et al. 2021).

Conventional plastic waste management strategies have proven inadequate, not merely due to processing limitations, but due to fundamental failures in collection and sorting infrastructure (Horodytska et al. 2022). Mechanical recycling, the dominant recovery method, faces significant challenges with heterogeneous and contaminated waste streams, often resulting in downcycled products of lower value rather than true circularity (Schyns and Shaver 2021). Consequently, the majority of waste is diverted to incineration or landfill. While incineration reduces volume, it releases significant GHG and secondary pollutants including dioxins, carbon monoxide, and nitrogen oxides, which pose severe environmental health risks (Bello et al. 2022). Similarly, landfilling, the most common disposal method, merely delays the problem, leading to a long-term accumulation of plastics that degrade exceedingly slowly (Wojnowska-Baryła et al. 2022).

The exponential growth in production coupled with these low recycling rates indicates a fundamentally unsustainable linear 'take-make-dispose' model. With global plastic waste projected to reach 1.2 billion tonnes by 2060, this trajectory compounds the current crisis of insufficient global processing capacity (UN Environment program 2025). This disparity highlights a fundamental market failure; the immediate economic gains from virgin plastic production

overshadow the immense, un-costed externalities associated with environmental remediation and public health. Therefore, the problem extends beyond simple management of plastic waste; it demands a shift toward a circular economy that reframes plastics not as end-of-life environmental liabilities, but as renewable feedstocks for new value chains.

1.2 Integrating molecular discovery with process engineering for a circular plastics economy

To enable this transition, we must move beyond a "one-size-fits-all" approach. This review distinguishes itself by integrating viewpoints often addressed separately, arguing that the solution to the "sorting and processing bottleneck" lies in emerging hybrid technologies facilitated by biological discovery. Two necessary outcomes for end-of-life plastics are distinguished:

1. **Resource recovery (Bio-recycling):** The controlled depolymerisation of plastics into monomers for re-manufacturing, preserving the embodied energy of the material.
2. **Environmental fail-safe (Mineralisation):** The complete degradation of escaped plastics into CO₂ and water to minimise ecological harm when recovery is impossible.

By integrating these viewpoints, this review bridges three critical domains to outline a holistic framework:

Molecular discovery (Section 2): We clarify the mechanistic basis of degradation and examine how omics, bioinformatics, and AI are uncovering novel enzymes capable of handling the heterogeneous waste streams that defeat current mechanical sorting.

Process engineering (Section 3): We critically evaluate the transition to next-generation recycling, specifically how biological discoveries are upscaled into hybrid systems, combining mechanical, chemical, and biological methods to create "tailored" recycling pathways.

Implementation strategy (Section 4): We propose a strategic roadmap based on four pillars material design, hybrid processing, adaptive pol-

icy, and digital logistics to align product design with engineering realities.

Ultimately, this review suggests that "design for circularity" and advanced hybrid processing must evolve concurrently. The amalgamation of molecular insight with process-scale innovations holds the key to decoupling the global plastic value chain from fossil feedstocks, transforming waste into a renewable resource for future manufacturing.

2 The biological toolkit for circularity: from mechanisms to discovery

To transition from a linear to a circular model, the biological mechanisms capable of dismantling synthetic polymers need to be harnessed. However, the application of these mechanisms depends entirely on the desired end-of-life outcome.

2.1 Mechanisms and determinants: mineralisation versus depolymerisation

Plastic degradation in the environment is typically initiated by abiotic processes, with exposure to light, heat, and acid/base conditions serving as the primary drivers of photodegradation, thermo-oxidative degradation, and hydrolysis, respectively (Gewert et al. 2015). These abiotic processes are often prerequisites for biodegradation, as they reduce the polymer's molecular weight and crystallinity, increasing bioavailability (Gewert et al. 2015). Following this, biotic degradation unfolds in specific sequential phases:

- (i) **Biodeterioration:** This initial phase involves the physical and chemical alteration of the plastic surface (Khoshtinat 2023). It is catalysed by diverse agents, ranging from higher order organisms such as waxworms, which mechanically damage plastic to facilitate chemical modification by their gut microbiota (Vital-Vilchis and Karunakaran 2025) to complex microbial consortia including bacteria, fungi, algae, and lichen (Zettler et al. 2013). These microorganisms form structured biofilms that inflict damage mechanically (e.g., *via* penetration and cracking), chemically (e.g., *via* secretion of

organic acids), and by initiating enzymatic attack (Yuan et al. 2020).

- (ii) **Bio-fragmentation:** This phase is driven by extracellular enzymes secreted by microorganisms (Ankita and Pandey 2025). Specific enzymes, such as oxidoreductases (including mono-oxygenases) and hydrolases (e.g., depolymerases), are responsible for fragmenting high molecular weight polymers into smaller molecules such as oligomers and monomers (Guajardo and Andler 2024). This reduction in size is critical as it enables these smaller molecules to pass through the microbial cell wall and cytoplasmic membranes for further metabolism.
- (iii) **Assimilation:** The subsequent phase is assimilation, wherein the fragments are taken up and metabolised by microbial catabolic pathways, channelling carbon and energy into cellular growth (Lucas et al. 2008).
- (iv) **Mineralisation:** Assimilated molecules may undergo mineralisation aided by catabolic pathways. There are three primary pathways: the aerobic pathway, which utilises O_2 ; the anoxic pathway, which employs inorganic species such as NO_3^- , SO_4^{2-} , Fe^{3+} , or S^0 ; and fermentative pathways, which use organic compounds as the

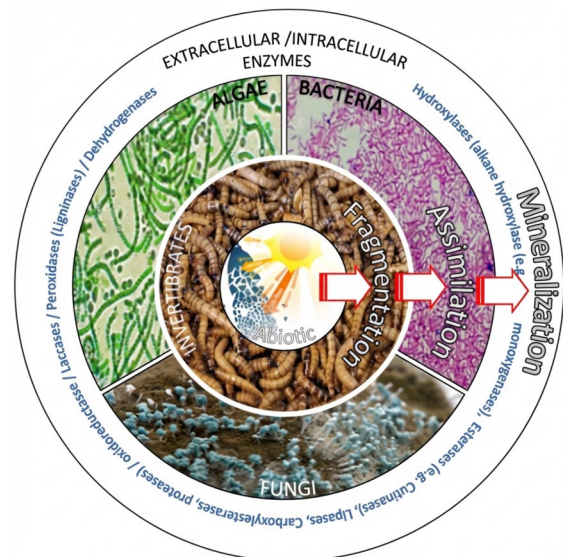


Fig. 2 Abiotic and biotic factors responsible for fragmentation, assimilation and mineralisation of plastics

final electron acceptors (Díaz 2004). A conceptual overview of these interrelated abiotic and biotic processes is depicted in Fig. 2.

While this full pathway describes the natural fate of polymers, applying it to a circular economy requires a strategic choice regarding where to stop the process. Therefore, a critical distinction must be made between allowing degradation to proceed to completion versus targeting specific intermediate stages:

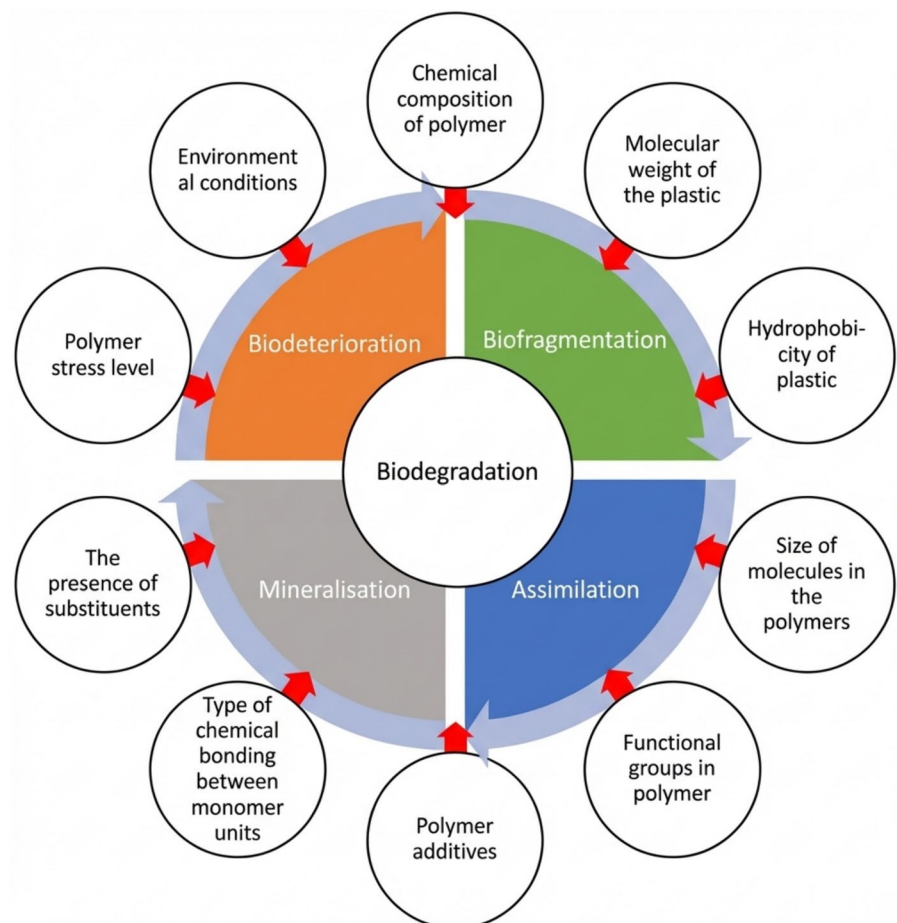
Mineralisation (Environmental fail-safe): This represents the complete biological breakdown of the material into simple compounds such as CO₂ and water. While this is the essential “harm reduction” mechanism for minimising the environmental accumulation of escaped plastic waste, relying on mineralisation for waste management effectively

wastes the energy and molecular value of these polymers.

Depolymerisation (Resource recovery): Conversely, the objective of circularity is to arrest the biological process at the bio-fragmentation stage. By utilising specific enzymes to hydrolyse polymers back into their constituent monomers (e.g., recovering terephthalic acid from polyethylene terephthalate (PET)) the material can be reclaimed for re-manufacturing. This approach effectively closes the loop, decoupling the plastic value chain from the need for continued fossil extraction.

The efficacy and rate of plastic degradation are not governed by a single factor, but by the complex interplay between the polymer’s intrinsic characteristics and extrinsic environmental conditions (Fig. 3) (Kliem et al. 2020).

Fig. 3 Factors affecting the biodegradation mechanisms of plastics (adapted from Artham et al. (2008))



2.1.1 Intrinsic determinants

A polymer's susceptibility to biodegradation is primarily determined by its chemical composition, molecular weight, hydrophobicity, and functional groups (Zambrano et al. 2020). Generally, polymers with heteroatom-containing backbones (e.g., polyesters, polyurethanes, polyamides), lower molecular weights, and amorphous structures are more accessible to microbial enzymes (Urbanek et al. 2020). Equally, densely cross-linked, non-oxygenated structures and highly ordered crystalline regions are particularly resistant to enzymatic attack (Zeikus 2018). This resistance is often compounded by additives; while pro-oxidants or starch can accelerate breakdown (Zheng et al. 2005), widespread additives such as plasticisers (e.g., phthalates), flame retardants (e.g., boric acid, brominated flame retardants), antioxidants, ultraviolet (UV) stabilisers (e.g., bisphenol A), heat stabilisers (e.g., cadmium and lead compounds), biocides (e.g., arsenic compounds), and pigments (e.g., cadmium compounds) notably impede natural degradation (Pritchard 1997).

2.1.2 Extrinsic drivers

Environmental conditions are equally critical. Factors such as pH, temperature, and nutrient availability (N, P) directly influence microbial activity and, consequently, degradation rates (Research et al. 2025). Crucially, degradation in real-world environments is rarely performed by lone organisms but by structured consortia (biofilms) (Shokunbi et al. 2025). Efficiency relies on the syntrophic activity of these communities, where different functional guilds act synergistically to physically damage the plastic and chemically metabolise intermediates (Shokunbi et al. 2025).

2.1.3 Engineering the interplay

Understanding the above relationships allows for active manipulation. For instance, abiotic pre-treatments such as heat or UV exposure are not merely passive precursors; they actively destabilise the polymer matrix by reducing crystallinity, introduce reactive functional groups to recalcitrant polymers like polyethylene (PE), and increase accessible surface area. This “primes” the material for biological action, demonstrating a direct cause-and-effect relationship

where intrinsic properties are favourably modified to overcome biological resistance (Brown et al. 2024). This complexity underscores why a “one-size-fits-all” solution is ineffective, necessitating a shift toward “design for biodegradability” where material properties are optimised for these known pathways from the outset.

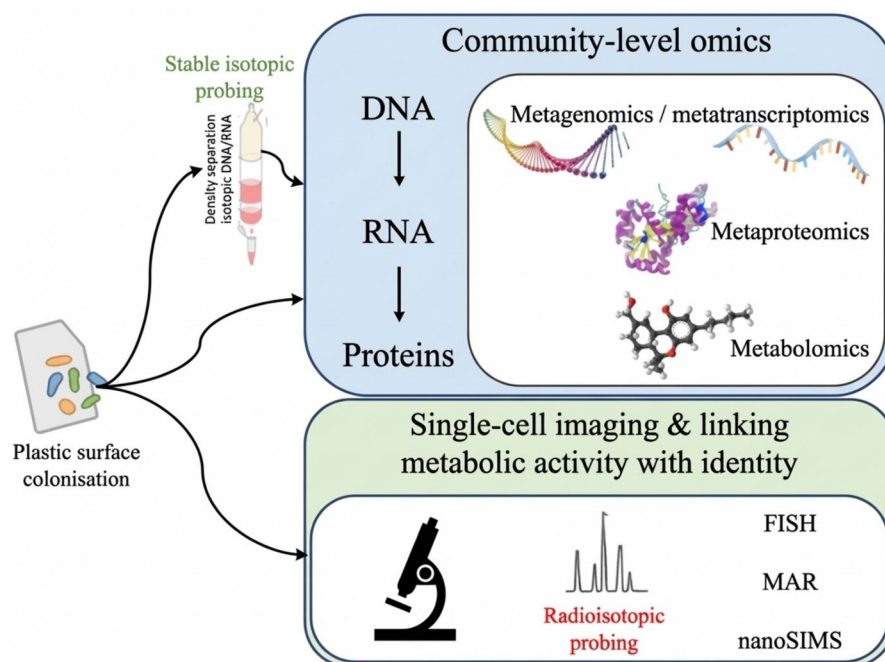
2.2 Mining the “Plastisphere” for industrial biocatalysts

A critical bottleneck in engineering bio-recycling systems is the limited availability of well-characterised microbial and enzymatic systems capable of efficiently depolymerising the vast diversity of polymer types. Traditional culture-based microbiology has yielded numerous isolates that can attack plastics, yet these represent only a small fraction of the complex communities (the “plastisphere”) found on or around plastic debris (Bai et al. 2024). Many plastic-degrading microbes are unculturable or exhibit their full degradative potential only in synergy with a community, meaning culture-dependent studies alone provide an incomplete view (Shilpa et al. 2022). To overcome this, researchers are now deploying an integrated suite of culture-independent tools (Fig. 4), spanning omics, advanced imaging, and bioinformatics to systematically mine nature for novel biocatalysts (Kim et al. 2022a).

2.2.1 Multi-omics insights into the plastisphere

Metagenomics has revolutionised the search for plastic-degrading organisms by bypassing the need for cultivation (Kim et al. 2022a). Among various metagenomic strategies (16S rRNA gene surveys, functional gene targeted sequencing, and whole-genome shotgun sequencing), shotgun metagenomics is particularly powerful for mining plastic-degrader genomes and genes (Kim et al. 2022a). This approach allows the reconstruction of genomes for uncultivated community members and for the identification of novel depolymerases, or catabolic enzymes encoded in those genomes. For example, a recent study utilised environmental metagenomics to discover several new candidate PET hydrolases from a compost microbiome, revealing an untapped diversity of enzymes active on PET that were not evident from cultured isolates (Sonnendecker et al. 2022).

Fig. 4 Phylogenetic and functional diversity of microbial communities with a combined use of molecular, chemical, isotope labelling and microscopy methods. *DNA* Deoxyribonucleic acid; *RNA* Ribonucleic acid; *FISH* Fluorescent in situ hybridisation; *MAR* Microautoradiography; *nanoSIMS* Nanoscale secondary ion mass spectrometry



To distinguish dormant genetic potential from active degradation, **metatranscriptomics** is employed to profile gene expression. This is critical because not all genes detected in a metagenome are functionally expressed under specific environmental conditions. Recent analyses have shown that when microbes colonise PE in landfill soil, genes encoding alkane monooxygenases (e.g., *alkB*) and PETases are significantly upregulated early in biofilm formation (MacLean et al. 2024). During later colonisation stages, transcripts for fatty acid β -oxidation pathways were highly upregulated, indicating active metabolism of PE breakdown products (MacLean et al. 2024). Notably, the expression of PETase genes in unpolluted forest soils suggests a wider, pre-existing capacity for polyester degradation than previously assumed (MacLean et al. 2024).

Complementary approaches such as **metaproteomics** (the mass-spectrometric identification of proteins in environmental samples) and **metabolomics** (profiling of small molecules) validate these findings by detecting the actual enzymes produced and the metabolic intermediates generated (Baran et al. 2009). For instance, if a cutinase or PETase enzyme is being produced by a microbe in a plastic-enriched microcosm, metaproteomics could confirm its presence and abundance, providing evidence of

an active degradation process. Metabolomics, on the other hand, can track small-molecule intermediates (e.g., dicarboxylic acids from PET or alkane oxidation products from PE) to infer active pathways. Together, these omics tools allow a multi-layered view: metagenomics tells us who might be capable of degrading plastic and what genes are available; metatranscriptomics and metaproteomics tell us which of those capabilities are actually deployed under given conditions; and metabolomics can often reveal the outcome of those activities in terms of chemical transformation. Such integrated meta-omic studies have highlighted that the complete degradation of recalcitrant polymers such as PE likely requires a consortium of organisms performing different steps, one species initiating oxidation of the polymer surface, others finishing oxidation and assimilating the breakdown products (MacLean et al. 2024).

2.2.2 Integrating omics and imaging for mechanistic insights into the plastisphere

While omics provide community-wide insights, they often lack spatial resolution and direct proof of metabolism (Moffitt et al. 2022). To validate

predictions in situ and link phylogenetic identity with metabolic function, high-resolution imaging and single-cell techniques are indispensable (Arias-Andres 2020).

2.2.2.1 Fluorescent *in situ* hybridisation (FISH) This technique uses fluorescent probes to visualise specific microbial taxa within biofilm architectures (Almeida and Azevedo 2021). When coupled with confocal laser microscopy, FISH enables the reconstruction of the three-dimensional (3D) spatial organisation of putative degraders on polymer surfaces, revealing features such as cell clustering, stratification of different taxa, and their proximity to the polymer interface (Bridier and Briandet 2014). However, FISH alone only reveals who is where, not what they are doing.

2.2.2.2 Isotope labelling and imaging (MAR and NanoSIMS) To confirm active metabolism, FISH is often combined with microautoradiography (MAR) or nanoscale secondary ion mass spectrometry (nanoSIMS). In MAR, cells metabolising radio-labelled plastic substrates (e.g., ^{14}C -labeled low-molecular-weight oligomers of a polymer) precipitate silver grains in a photographic emulsion, distinguishing active degraders from bystanders (Ginige 2005). While MAR powerfully validates activity, it is only semi-quantitative and is limited by the resolution of the emulsion (Lund Nielsen and Halkjær Nielsen 2005). NanoSIMS provides superior precision, allowing for the imaging of elemental and isotopic composition at sub-micron resolution (Gyngard and Steinhäuser 2019). This enables researchers to trace the incorporation of stable isotopes (e.g., ^{13}C -labeled polymer or a key intermediate such as ^{13}C -monohydroxyethylene terephthalate (MHET) for PET degraders) into individual cells. Recent studies combining scanning electron microscopy (SEM), FISH, and nanoSIMS visually confirmed that specific bacterial cells on PE and PS surfaces were assimilating carbon derived from additives (Xiang et al. 2024).

2.2.2.3 Nucleic acid-based stable isotope probing (DNA-SIP and RNA-SIP) Complementing imaging approaches, molecular SIP bridges community omics with function by tracing isotopic flow directly into genetic material. In this workflow, a microbial community is fed ^{13}C -labeled substrates, after which DNA or RNA is extracted and density-separated into

“heavy” (^{13}C -incorporated) and “light” nucleic acid fractions (Dumont and García 2020). Sequencing the “heavy” fraction selectively identifies the microbes that grew using the labelled substrate. This method has been applied, for instance, to identify soil microbes assimilating ^{13}C -labeled polyethylene glycol, providing definitive genetic evidence of polymer utilisation (Goudriaan et al. 2023).

2.2.2.4 Confocal Raman microscopy (CRM) Unlike conventional methods that often require destructive sampling, CRM is a non-destructive, label-free technique that utilises inelastic light scattering to provide a unique chemical fingerprint of the sample (Schlücker 2012). This water-independent capability is particularly advantageous for biological systems, enabling in situ, real-time monitoring of biodegradation with speed and low operational costs (He et al. 2024). CRM uniquely integrates three critical analytical dimensions: identification (discriminating between microorganisms and characterising polymer chemical bonds), depth visualisation (enabling 3D mapping of biofilm architecture), and quantification (measuring the extent of polymer breakdown). This versatility supports the high-throughput screening of potential degraders and allows for the detection of microplastics as small as 1 μm , revealing detailed insights into their chemical composition and structural changes during degradation (Phan et al. 2022). Furthermore, by monitoring specific metrics such as the decrease in film crystallinity, CRM provides a direct, quantitative measure of degradation progression (Urban 1989).

The integration of meta-omics with high-resolution imaging and single-cell techniques constitutes a comprehensive toolbox for elucidating plastic biodegradation. While omics approaches reveal the genetic potential and transcriptional status of microbial communities, techniques such as FISH, MAR, nanoSIMS, SIP, and CRM bridge the gap between genetic potential and phenotypic reality. By linking microbial identity to specific metabolic activities and quantifying the resulting structural changes in the polymer, these tools enable the rigorous, mechanistic understanding required to translate natural discoveries into engineered applications.

2.3 Accelerating optimisation with bioinformatics and artificial intelligence (AI)

The massive datasets generated by multi-omics approaches necessitate robust computational platforms to translate raw sequence data into functional biological insights. Over the past decade, the convergence of curated databases, structural bioinformatics, and artificial intelligence (AI) has created a “discovery engine” capable of identifying and optimising biocatalysts at a pace previously unattainable.

2.3.1 Functional annotation and pathway prediction

The first step in mining the plastisphere is connecting genomic sequences to metabolic potential. Foundational resources like **enviPath** (Environmental Contaminant Biotransformation Pathway Resource, available at <https://envipath.org/>) database (the successor to EAWAG-BBD) catalogue metabolic pathways for a wide range of xenobiotics, including plastics-related chemicals, and provides information on the organisms and enzymes involved (Wicker et al. 2015) allowing researchers to predict likely degradation routes for polymer intermediates based on known biotransformation rules.

More specialised resources are critical for plastic-specific discovery. The Plastics Microbial Biodegradation Database (PMBD) contains curated data on over 900 microorganism-plastic relationships and thousands (> 8,000) of enzyme sequences predicted to effect plastic degradation (Gan and Zhang 2019). By querying these repositories, researchers can rapidly assess whether recovered metagenomic sequences encode genes for known degraders (e.g., PETases, polyurethane (PU) esterases) or possess conserved catalytic motifs indicative of degradation potential. Specialised databases such as Organonitrogen Degradation Database (ONDB, for urethanes), OxDBase (for oxygenases), AromoDeg (for aromatic compound degradation (relevant to polystyrene or PET byproducts)) and RHObase (focusing on ring-hydroxylating dioxygenases) can further refine this search, helping to identify enzymes capable of attacking specific polymer backbones or additives (Arora et al. 2022).

When integrated with genome annotation pipelines (e.g., rapid annotation using subsystems (RAST), Kyoto Encyclopedia of Genes and Genomes

(KEGG)), these tools allow for the reconstruction of complete metabolic pathways, distinguishing organisms that can merely fragment plastic from those capable of full assimilation. For instance, following genome sequencing of a novel strain, one can rapidly check for the presence of pathways containing genes encoding for depolymerases for polymers such as polyesters (e.g. PET hydrolases for PET), those involved in the protocatechuate pathway for breaking down the terephthalate monomer (a major breakdown product of PET), or those involved in alkane degradation pathways for plastics such as PE.

While bioinformatic prediction alone does not constitute proof of function, it significantly narrows the search space. For example, *Ideonella sakaiensis* was found to have a PETase and MHETase (which breaks down mono(2-hydroxyethyl) terephthalate) through traditional biochemical screening (Yoshida et al. 2021). With the growing availability of genome sequences from environmental isolates, homologues of these enzymes can now be systematically identified, enabling the discovery of additional species with potential PET-degrading capabilities.

2.3.2 Structural bioinformatics and in silico assessment

Sequence similarity alone is often insufficient to predict activity on recalcitrant polymers. To bridge the gap between sequence and function, advanced structural bioinformatics tools are employed to simulate enzyme–substrate interactions. For example, AlphaFold2 can predict a novel enzyme’s 3D structure (Yang et al. 2023); docking algorithms can then explore its polymer binding capabilities (Parida et al. 2024); and cutting-edge AI-based prediction tools such as AlphaFold3 (Abramson et al. 2024) and Boltz2 (Passaro et al. 2025) can simultaneously predict both protein and ligand structures as complexes. Boltz2 further enhances this by providing quantitative binding affinity scores, while tools such as P2Rank can identify active site features suited for polymer cleavage (Krivák and Hoksza 2018), all of which are invaluable for guiding experimental priorities. Pathway prediction systems can map routes from polymer breakdown products (e.g., alkanes from polypropylene) or additive leachates to central metabolites, informing analytical targets and genetic experiments

(Wicker et al. 2010). By integrating curated databases (e.g., PMBD) with predictive models, these approaches connect genes to pathways, accelerating discovery and moving toward comprehensive *in silico* screening of environmental datasets for novel plastic-degrading enzymes and organisms.

2.3.3 From functional prediction to generative design via AI

The most transformative shift in the biological toolkit is the application of Machine Learning (ML) and AI to sift through massive datasets, from genomics and proteomics to scientific literature to pinpoint promising biocatalysts for plastic degradation (Medina-Ortiz et al. 2025; Zahra et al. 2025). This convergence of big data and algorithms is enabling a shift from unexpected discovery to systematic design (Bergeson et al. 2024).

2.3.3.1 Predictive classification with protein language models To enable high-throughput discovery, ML algorithms are trained to recognise sequence or structural patterns that define plastic-degrading enzymes. For instance, researchers have compiled datasets of enzymes known to act on plastics (such as PET hydrolases, polylactic acid (PLA) depolymerases, and PE oxygenases) and used these to train classification models (Jiang et al. 2023; Medina-Ortiz et al. 2025). Advanced protein language models capture subtle motifs in enzyme sequences (Tran et al. 2023), creating “embeddings” that serve as sophisticated fingerprints for function. Researchers have successfully integrated these pre-trained models with ML classifiers to screen vast gene libraries (Medina-Ortiz et al. 2025). In a recent framework, binary classification models achieved approximately 89% accuracy in identifying enzymes targeting specific plastics such as PET, PLA, and nylon (Medina-Ortiz et al. 2025). When applied to public databases, this approach filtered over 6,000 sequences down to a manageable list of high-probability candidates, drastically narrowing the experimental search space from thousands of random tests to a few dozen prioritised targets.

2.3.3.2 Generative optimisation and *in silico* evolution Beyond classifying existing sequences, generative AI models are now being used to design completely new enzyme variants (Xie and Warshel

2023). By combining generative models with functional classifiers, researchers can evolve enzymes *in silico*, proposing mutations that tailor them for specific industrial conditions (Medina-Ortiz et al. 2025). For example, AI-generated PETase variants have been modelled with distinct properties, such as lower molecular weight and higher aliphatic indices, which are potentially favourable for interacting with hydrophobic plastic surfaces (Medina-Ortiz et al. 2025).

A landmark achievement validating this computational approach is the development of FAST-PETase, a remarkably efficient PET-degrading enzyme (Lu et al. 2021). Starting with a natural PETase, researchers employed an ML-guided directed evolution strategy to identify mutations that would enhance thermostability. By training a model on mutant libraries, the algorithm predicted the amino acid substitutions most likely to improve performance. The resulting engineered enzyme can depolymerise PET at 50 °C notably faster than its wild-type counterpart and near the polymer’s glass transition temperature (Lu et al. 2021). This success demonstrates the power of ML to streamline optimisation, condensing tasks that would require many rounds of trial-and-error mutation into a targeted, data-driven process.

2.3.3.3 Microbial phenotype prediction ML is also accelerating microbial discovery by analysing large metagenomic datasets. Unsupervised learning can cluster samples to highlight “hotspots” of degradation activity (Ghannam and Techtmann 2021), while emerging predictive models can infer an organism’s specific degradative phenotype (e.g., “polyurethane degrading”) directly from its genome (Koblitz et al. 2025). This allows researchers to scan thousands of isolates and flag those possessing the specific genomic hallmarks of a plastic degrader for follow-up testing.

Collectively, the convergence of multi-omics discovery with AI-driven functional prediction and generative design has advanced the field from largely empirical discovery to systematic engineering, yielding a process-ready portfolio of industrial biocatalysts tailored for industrial operating conditions. Identification of an effective biocatalyst is a necessary but insufficient condition for impact. Molecular-level optimisation must be coupled to scalable process

design to achieve practical relevance. Accordingly, the following section focuses on the integration of molecular-scale advances into hybrid recycling systems, with emphasis on overcoming the economic, kinetic, and logistical constraints that govern global waste management.

3 Process engineering: from mechanical to hybrid systems

To operationalise the biological toolkit described in the previous section, it must be benchmarked against and integrated with existing industrial frameworks. This section critically evaluates the evolving recycling landscape (Fig. 5), transitioning from the thermodynamic limitations of conventional mechanical reprocessing to the techno-economic realities of next-generation hybrid systems.

3.1 Mechanical recycling (Reprocessing)

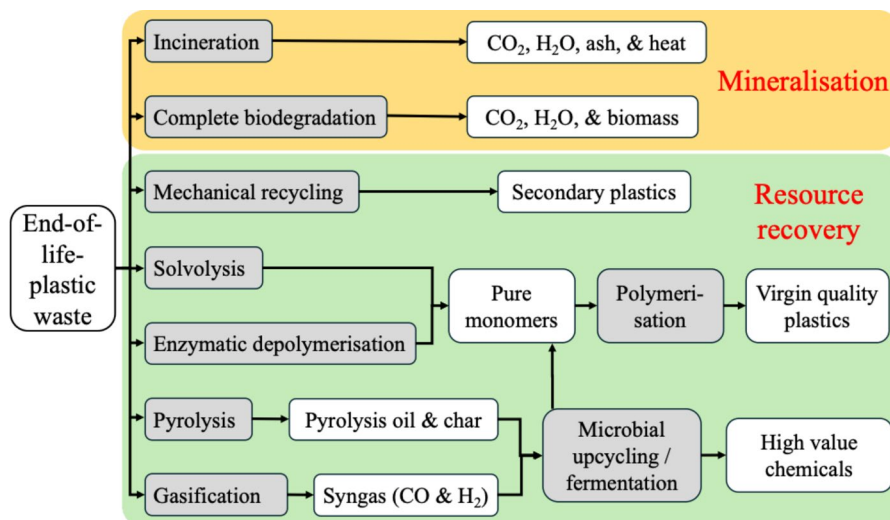
Mechanical recycling remains the dominant recovery method, accounting for more than 90% of all global plastic recycling activity (Chen and Hu 2024). This process, which involves collection, sorting, cleaning, shredding, and re-pelletising, is a straightforward and low-cost approach that preserves the polymer's chemical structure by employing physical rather than chemical methods (Schade et al. 2024). Consequently, it is thermodynamically

efficient, significantly reducing energy use and GHG emissions compared to both virgin plastic production and emerging chemical recycling technologies (Shan et al. 2023). Its success is most evident in established programs for polyethylene terephthalate (PET) bottles and high-density polyethylene (HDPE) containers, which achieve the highest recovery rates (Hadjilambrinos 1996) and support secondary markets, such as the conversion of PET bottles into polyester fibres (Birla et al. 2025).

However, as noted in the introduction, the process has a fundamental limitation; its dependence on homogeneous feedstocks (Lahl and Zeschmar-Lahl 2024). Mechanical recycling is highly sensitive to contamination from food residues, labels, and non-plastic materials, necessitating extensive and costly sorting (Schyns and Shaver 2021). This logistical challenge is compounded by modern packaging trends, particularly complex, multi-layered designs and diverse additives that are neither easily separable nor recyclable together (Ciawi et al. 2025).

Furthermore, the process is limited by material degradation. During reprocessing, plastics undergo thermal and oxidative stress, leading to polymer chain scission (Schyns and Shaver 2021). This results in “downcycling”, a progressive deterioration of mechanical and thermal properties that restricts the number of recycling cycles and limits applications to lower-value products (Yao et al. 2013). Because of these constraints, mechanical

Fig. 5 Strategic pathways for end-of-life plastic management



recycling currently handles less than 20% of total global plastic waste, with the vast remainder diverted to landfill or incineration (Nolan 2023). Thus, while mechanical recycling remains an essential tool for relatively pure waste streams, it cannot fundamentally resolve the challenge posed by complex, mixed plastic waste.

3.2 Chemical recycling (Thermochemical and solvolysis)

Chemical recycling employs heat, chemical agents, or catalysts to depolymerise plastic waste into monomers or petrochemical feedstocks (Chen et al. 2021). Unlike mechanical methods that retain the polymer backbone, chemical recycling breaks the material down to its molecular constituents, which can be re-polymerised or converted into fuels (Datta and Kopczyńska 2016). This approach is strategically complementary to mechanical recycling, as it can process soiled, mixed, and multilayer plastics that would otherwise be landfilled (Schade et al. 2024). Broadly, these technologies are divided into thermochemical processes and solvolysis (depolymerisation in solvents).

Thermochemical processes include pyrolysis and gasification. These high-temperature methods are primarily utilised for mixed waste streams, particularly polyolefins (non-oxygen containing PE, PP) that are difficult to separate.

3.2.1 Pyrolysis (400–600 °C)

This process decomposes plastics in the absence of oxygen to produce liquid oil (such as TACOIL produced by Plastic Energy), gases, and char (Miandad et al. 2016). It offers significant feedstock flexibility and is currently being scaled up by companies such as Brightmark (USA) and Plastic Energy (UK/Europe) (Tullo 2019, 2020). However, pyrolysis faces strict chemical limitations: polymers containing heteroatoms (e.g., oxygen in PET and chlorine in polyvinyl chloride (PVC)) generate hazardous by-products (such as HCl and char containing heavy metals or polyaromatic hydrocarbons (PAHs)) and degrade oil quality, necessitating costly downstream refining (Terrell and Garcia-Perez 2021).

3.2.2 Gasification (700–1000 °C)

Operating at higher temperatures with partial oxidation, gasification converts plastics and municipal solid waste into syngas (CO, H₂), which can be used for energy or as a feedstock for fuels and chemicals (Lee et al. 2014). While widely applied to biomass (Klein and Themelis 2003), its application to plastic is complex due to feedstock variability, complicated reaction mechanisms, and multi-phase transport phenomena (Madanikashani et al. 2022). Nevertheless, it achieves substantial volume reduction (80–90%) (Yepes Maya et al. 2016) and effectively destroys hazardous organic compounds (Asikainen et al. 2002), making it a promising route for hydrogen production or industrial fuel generation, as demonstrated in waste-to-energy plants in Japan and Europe (Leckner 2015).

Despite their robustness against contamination, thermochemical processes are inherently energy intensive. Critics note that without rigorous Life Cycle Assessment (LCA), the carbon footprint of heating plastics to these temperatures can exceed that of virgin plastic production, particularly if process efficiency is low or if extensive purification is required to remove heavy metals and halogens from the output.

3.2.3 Solvolysis (Depolymerization)

Solvolysis uses solvents and reagents to break down condensation polymers (such as PET, polyamides, and polyurethanes) at lower temperatures than pyrolysis (Jiang et al. 2022). Unlike thermochemical methods which produce a mix of hydrocarbons, solvolysis targets the recovery of specific monomers with high purity.

Solvolysis can achieve monomer yields exceeding 90%, producing “virgin-quality” outputs suitable for closed-loop manufacturing. This is demonstrated by commercial-scale operations from Eastman and Carbios, which recycle PET waste into new resin (McNeeley and Liu 2024). While solvolysis avoids combustion-related emissions, it requires substantial solvent inputs and significant energy for monomer separation and purification (Keith et al. 2025). Its economic viability is therefore tightly coupled to the market value of the recovered monomers relative to recovery costs. Consequently, solvolysis is best suited

to high-value, relatively pure waste streams (such as textiles and bottles) rather than serving as a universal solution for mixed plastic waste.

3.3 Bio-recycling: industrialising the "Plastisphere"

This section addresses the application of the biological toolkit described in Sect. 2. In contrast to the physical force inherent to thermal processes, biological recycling proceeds under mild conditions (ambient temperature and pressure), offering a potentially lower-carbon pathway for the recovery of value from specific polymer classes (Jiao et al. 2021).

3.3.1 Enzymatic systems

The core opportunity of biological recycling lies in the precision of enzymatic catalysis. Unlike thermal processes that indiscriminately cleave bonds, enzymes function as "molecular scissors", targeting specific chemical linkages under mild conditions (Herbert et al. 2022). A paradigm shift occurred with the discovery of the bacterium *Ideonella sakaiensis*, which utilises a two-enzyme system (PETase and MHETase) to hydrolyse PET into its constituent monomers, terephthalic acid and ethylene glycol (Kramer et al. 2024). This discovery spurred a wave of protein engineering; for instance, the development of FAST-PETase, a variant engineered *via* ML, has overcome natural kinetic limitations, achieving > 90% depolymerisation of PET in under 48 h at 50 °C (Lu et al. 2021). Beyond PET, the biological portfolio is expanding to include esterases for polyurethanes (PU) and amidases for polyamides (Magnin et al. 2019), offering a low-energy pathway to recover high-purity monomers from polymers that are historically difficult to recycle chemically.

Driven by these molecular advances, the translation of enzymes from the lab to industrial operation is now commercially active. Companies such as Carbios (France) and Samsara Eco (Australia) are operationalising these biocatalysts in large-scale bioreactors, targeting annual processing capacities of 50,000 tonnes and 1.5 Mt, respectively by 2030. The definitive advantage of this scaling is selectivity; because enzymes specifically attack the polymer backbone,

they can process complex, multi-material wastes such as opaque bottles and polyester-cotton blends that traditional mechanical and thermal methods cannot handle (Berdugo-Clavijo et al. 2022). The sector is further accelerating through AI-driven engineering by players such as Birch Biosciences (USA) and Enzymity (Latvia), while Plasticentropy (Spain) is broadening the scope by exploring novel enzymes from wax moth saliva for the oxidation of polyolefins (Spínola-Amilibia et al. 2023).

3.3.2 Microbial consortia and whole-cell systems

While purified enzymes offer precision, whole-cell microbial systems provide metabolic versatility (Carballeira et al. 2009). In nature, complete plastic mineralisation is limited by slow kinetics, often taking centuries (Mohan et al. 2020). However, in engineered bioreactors, these rates can be dramatically accelerated using enriched microbial consortia (Danco et al. 2019). These mixed communities offer a robust "division of labour": distinct members act synergistically, with some secreting depolymerases to fragment the polymer, while others metabolise the resulting oligomers (Shokunbi et al. 2025). This approach is the frontier of synthetic biology, where researchers are now co-culturing engineered microbes to move beyond mere degradation (mineralisation to CO₂) towards upcycling. In these systems, one strain depolymerises the plastic waste, while a partner strain ferments the monomers into high value bioproducts such as biofuels or bioplastics, effectively closing the carbon loop (Bao et al. 2023).

3.3.3 Bioreactor configurations for solid waste treatment

To maximise the kinetics of these whole-cell and enzymatic processes, reactor designs are being adapted from wastewater treatment to handle solid plastic feedstocks. The choice of configuration is strictly governed by the polymer's physical properties (e.g., hydrophobicity) and the target end-product:

Biofilm reactors (Fixed-bed and moving bed biofilm reactors (MBBRs)): These are critical for hydrophobic plastics such as PE and PP (Shabbir et al. 2020). Because these polymers repel water, the rate-limiting step is the contact between the

microbe and the plastic surface. MBBRs promote surface-attached growth, ensuring high-density colonisation and prolonged contact time essential for attacking the C–C backbone (Benes et al. 2025).

Membrane bioreactors (MBRs): MBRs utilise filtration systems to decouple hydraulic retention time from solids retention time (Kraume et al. 2005). This allows for the retention of specialised, slow-growing microorganisms that would otherwise be washed out of a continuous system, significantly enhancing the degradation of recalcitrant compounds (Sengupta et al. 2022).

Solid-state composting reactors and anaerobic systems: For compostable plastics such as PLA, solid-state fermentation under thermophilic conditions has proven highly efficient (Itävaara et al. 2002). Conversely, while standard anaerobic digesters struggle with petroleum plastics, they show promise for bioplastics or pre-treated streams, where they can be integrated to recover energy *via* biomethane production (Nachod et al. 2021).

Continuous-flow systems: Moving beyond simple batch processing (limited by nutrient depletion), continuous-flow reactors allow for steady-state operation (Europa et al. 2000). These have been successfully demonstrated for the complete degradation of polyhydroxybutyrate (PHB) and hold potential for industrial-scale co-digestion with activated sludge (Lera et al. 2024).

3.3.4 Techno-economic analysis (TEA) and life cycle assessment (LCA) reality check

Despite the appeal of operating under “mild conditions,” biological recycling may not always be intrinsically sustainable (Woodley 2022). Rigorous LCAs indicate that enzyme production is itself energy- and resource-intensive, requiring fermentation, downstream purification, and substantial water inputs (Hobusch et al. 2024). Recent TEAs further suggest that, for enzymatic recycling to be economically competitive with virgin PET, enzyme yields must increase significantly, and operational lifetimes must be extended substantially (Uekert et al. 2023). In addition, downstream processing steps, particularly the separation and purification of

monomers from dilute aqueous streams can incur significant energy penalties through distillation, crystallisation, or related unit operations (Janković et al. 2024). Consequently, bio-recycling should not be regarded as a “magic bullet,” but rather as a specialised, high-selectivity approach best suited for challenging feedstocks (e.g., mixed or textile-derived polymers) for which mechanical recycling is ineffective or impractical.

3.4 Photochemical degradation

In the natural environment, photodegradation driven by solar UV-A and UV-B radiation is the primary mechanism of plastic weathering (Andrady 2015). However, this uncontrolled process takes centuries to degrade polyethylene films or bottles and is detrimental, as it leads to the fragmentation of macroplastics into persistent secondary microplastics rather than complete mineralisation (Kubowicz and Booth 2017).

In a process engineering context, however, this mechanism is deliberately accelerated and harnessed not as a disposal method but as a strategic pre-treatment tool. By controlling exposure conditions and introducing catalysts, the degradation timelines are compressed from decades to hours, and this is realised incorporating three key technological applications:

Pro-degradant additives: Incorporating metal stearates (e.g., iron or cobalt compounds) into the polymer matrix generates free radicals under UV or thermal exposure (Zheng et al. 2005). This initiates rapid chain scission, causing the plastic to become brittle and fragment, facilitating easier mechanical processing.

Photocatalytic nanocatalysis: The integration of semiconductor nanoparticles, such as titanium dioxide (TiO₂) or zinc oxide (ZnO), harnesses UV light to generate highly Reactive Oxygen Species (ROS), including hydroxyl radicals and singlet oxygen (Lee and Li 2021). These species aggressively attack the polymer backbone, accelerating breakdown without the need for harsh external chemicals.

Surface functionalisation (Pre-treatment): Most critically for hybrid systems, UV treatment oxidises the inert plastic surface, introducing oxygen-containing functional groups (carbonyls and hydroxyls). This chemical modification transforms

the surface from hydrophobic to hydrophilic, drastically increasing its affinity for water and microbial colonisation (Taghavi et al. 2021).

Industrial photodegradation is rarely employed as a standalone endpoint. While it is effective at cleaving polymer chains, it carries a significant risk of generating microplastic litter if the resulting fragments are not immediately contained and assimilated. Consequently, its optimal role is as a chemical facilitator that primes recalcitrant materials for subsequent biological attack. This synergy where photonic oxidation prepares the substrate for microbial mineralisation forms the basis of the intimate coupling of photochemical and biodegradation (ICPB) systems described in the hybrid frameworks below.

3.5 Hybrid recycling pathways: the integrated frameworks

Dealing with the vast diversity and volume of plastic waste requires a multifaceted approach, as no single technology is effective in isolation. The most promising solutions lie in hybrid processes that systematically integrate multiple methods to leverage their respective strengths using physical sorting for bulk efficiency, chemical processes for recalcitrant fractions, and biological systems for specificity and upcycling (Chakma and Moholkar 2014).

3.5.1 Mechanical + thermochemical (*The industrial baseline*)

This integration is already emerging in practice. In this cascaded model, high-quality streams (clean PET/HDPE) are sorted for mechanical reprocessing to preserve embodied energy. The “rejected” remainder consisting of dirty, mixed, or multi-layered plastics is diverted to chemical recycling units (pyrolysis or gasification). By doing so, facilities maximise material recovery while utilising energy-intensive chemical methods only as a “last resort” for fractions that mechanical recycling cannot handle.

3.5.2 Chemo-bio hybrids (*The frontier of circularity*)

These systems couple abiotic breakdown with biological valorisation. Here, a chemical or photochemical

step reduces the polymer’s molecular weight or alters its surface, creating accessible substrates for microbial upcycling.

3.5.3 Bio-upcycling

Rather than simple mineralisation, the goal is to convert waste into high-value biochemicals. For example, pyrolysis oil can be fed to oleaginous yeast to produce microbial lipids or biodegradable PHAs (Mihreteab et al. 2021), effectively transforming fossil-based polypropylene into bio-plastic precursors.

3.5.4 Enzymatic pre-treatment

Oxidative enzymes such as laccases can be used to graft functional groups onto inert surfaces (e.g., PE or polycarbonate), increasing hydrophilicity and enhancing subsequent biofilm-mediated degradation (Ray et al. 2023).

3.5.5 Intimate coupling of photocatalysis and biodegradation (ICPB)

This advanced configuration integrates photocatalytic oxidation and biological treatment within a single reactor (Yu et al. 2020). A key challenge is protecting microbial communities from UV radiation and reactive oxygen species generated by the photocatalyst. This is addressed through the use of macroporous carriers that support biofilm formation, shielding microorganisms from oxidative stress while permitting the diffusion of plastic-derived oligomers (Marsolek et al. 2008; Yu et al. 2020). Technologies such as the Photocatalytic Circulating-Bed Biofilm Reactor (PCBBR) have successfully demonstrated this coupling (Marsolek et al. 2008). Furthermore, next-generation Visible-Light Photocatalysis Coupled with Biodegradation (VPCB) systems have outperformed UV-based counterparts (Zhou et al. 2015), offering the potential to harness solar energy to continuously fragment plastic particles while a microbial consortium metabolises the released products.

ICPB systems can be further integrated with bioelectrochemical systems (BES), such as microbial fuel cells (MFCs) or microbial electrolysis cells (MECs). In such configurations, microbes oxidising plastic-derived organic compounds at the BES anode could

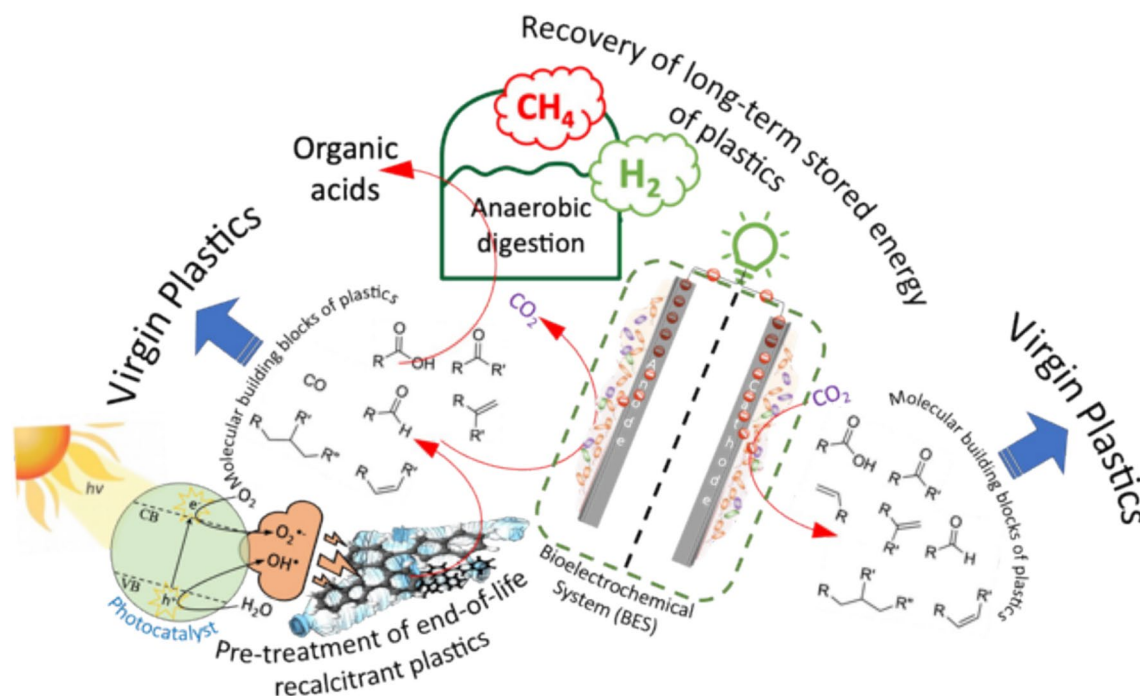


Fig. 6 A schematic illustration of the intimate coupling of photocatalysis with a bioelectrochemical system and/or anaerobic digester to recover chemical energy (as electricity,

hydrogen or methane) from end-of-life plastic and facilitate the production of new plastics through synthesis of molecular building blocks of plastics

generate electrical current (Lovley 2006), while the application of an external voltage could, conversely, enhance microbial degradation by facilitating better electron transfer (Chun et al. 2013). Moreover, electrons derived from plastic oxidation can be channelled to the BES cathode to drive microbial electrosynthesis of plastic precursors and other molecular building blocks (Kim et al. 2022b). Similarly, coupling ICPB with anaerobic digestion also may provide synergistic benefits through enhanced resource recovery and energy generation (Fig. 6).

3.5.6 Integrated industrial ecology (Recycling parks)

Beyond laboratory studies, hybrid processes are increasingly being implemented at the facility scale. Nations such as Sweden are pioneering integrated recycling parks, where mechanical sorting plants, chemical recycling units, and waste-to-energy facilities are co-located. This enables a

comprehensive approach, in which waste streams are dynamically routed to the most appropriate process based on material quality, thereby maximising overall recycling rates and reducing landfill diversion to near zero.

The future of plastic management lies not in choosing between mechanical, chemical, or biological methods, but in the strategic integration of all three. “Next generation” recycling is defined by the ability to route every fraction of a mixed waste stream to its optimal processing technology, a logistical challenge addressed in the final roadmap.

4 Roadmap for a global, integrated plastics economy

The scientific and engineering advances detailed in Sects. 2 and 3 provide the building blocks for a circular economy; however, they do not, in themselves,

Table 1 Strategic roadmap for end-of-life plastic management

Focus area	Categorisation	Technology	Standards	Logistics
Current State	Classification by resin code (e.g., polyethylene terephthalate (PET), polypropylene (PP)) often ignores additives and degradation potential	Over reliance on mechanical recycling; limited scalable solutions for mixed/soiled waste streams	Fragmented International Organisation for Standardisation (ISO)/ASTM standards focused on lab-scale biodegradability	Fragmented, co-mingled collection systems characterised by inefficient aggregation and high feedstock contamination
Desired Future State	Classification by degradation resistance: Categories based on susceptibility to mechanical, chemical, or biological breakdown	Tailored hybrid pathways: Waste streams routed to optimal Mechanical, Chemical, or Bio-systems	End-of-Life verification: Harmonised standards validating performance in industrial conditions	Smart infrastructure: Digitally integrated reverse supply chains leveraging artificial intelligence (AI) for autonomous sorting and decentralised material recovery
Strategic priorities	Audit additive toxicity	Advanced sorting and pre-treatment	Product-specific recovery protocols	Standardised bin systems
Short term	Re-classify based on hybrid processing suitability	Scale chemo-bio hybrid pilots	Certification for hybrid process compatibility	Digital tracking of waste flows
Medium term	“Design for degradation” mandates	Integrated industrial recycling parks	Global digital product passports	Autonomous collection and localised processing
Long term				

constitute a system. At present, global plastic waste management remains fragmented, relying on a disconnect between upstream material design and downstream processing infrastructure (Erdle and Eriksen 2023). This vulnerability was exposed by the collapse of the global waste trade following China's National Sword policy in 2017 (Brooks et al. 2018), which forced many nations to confront the inadequacy of their domestic capabilities.

To transition from theoretical circularity to practical implementation, a strategic realignment is required. The roadmap in Table 1 outlines four critical focus areas: material design, hybrid processing, adaptive policy, and digital logistics, bridging the gap between molecular-level innovation and global waste management systems.

4.1 Material design (Categorisation): from polymer type to degradation resistance

Current classification systems, such as the Resin Identification Codes (1–7), represent a significant barrier to circularity. Inherently manufacturing-centric, they categorise plastics solely by chemical backbone, failing to account for critical structural factors such as crystallinity, cross-linking, and the presence of more than 10,000 distinct chemical additives that ultimately govern end-of-life fate (Law and Narayan 2022). This creates a fundamental disconnect between production and disposal; for example, a mono-material PET bottle and a multilayer, additive-rich PET textile are classified identically, despite requiring radically different treatment pathways. Consequently, stakeholders across the value chain are left with oversimplified data that fails to reliably predict real-world degradability and processability.

4.1.1 The strategic shift

To resolve this gap, we propose shifting to a classification system based on degradation resistance. Rather than defining a material by its origin, this holistic model categorises plastics by their specific susceptibility to: Mechanical fragmentation (suitable for wash-and-melt reprocessing); Abiotic oxidation (susceptible to chemical or photochemical pretreatment (e.g., for ICPB)) and Biotic mineralisation (susceptible to enzymatic hydrolysis or microbial mineralisation).

This new framework necessitates a fundamental “**Design for Degradation**” approach. By explicitly linking classification to these material properties, manufacturers are mandated to align product design with viable infrastructure from the point of synthesis. For example, single-use packaging with high food contamination risk would be explicitly designed for the biotic mineralisation category, requiring the formulation to be free of microbial inhibitors to ensure compatibility with the bio-recycling systems described in Sect. 3.3. This creates a transparent, actionable link between a product and its optimal end-of-life pathway, replacing theoretical recyclability with verified process compatibility.

4.2 Hybrid processing (Technology): scalable resource recovery

The failure of the current circular economy is fundamentally a failure of alignment. While the necessary biological processes and engineering frameworks exist (as detailed in Sects. 2 and 3), the uncontrolled diversity of commercial plastics renders these solutions economically unscalable. Current waste streams are too heterogeneous for efficient mechanical processing, yet too low in value to justify the capital investment required for standalone chemical recycling plants (Schade et al. 2024). This systemic misalignment converts what should be a recoverable resource into a persistent environmental liability.

4.2.1 The strategic shift

The transition from disposal to asset recovery, driven by the Design for Degradation model, does not depend on creating a niche process for every unique plastic product. Instead, it requires the development of robust, versatile process engineering solutions tailored to the broad degradation categories defined above. This paradigm shifts the objective from simple disposal (mineralisation) to active resource recovery, enabling the extraction of high-value monomers and platform chemicals. Success hinges on the intelligent integration of existing knowledge rather than relying solely on the invention of entirely new technologies. It requires the strategic harmonisation of established

mechanical, abiotic, and biotic processes to manage streamlined input streams.

To bridge the gap between heterogeneous waste streams and scalable processing technologies, a staged, multifaceted strategy is required. Focus areas for short-, medium- and long-term efforts are outlined below.

4.2.1.1 Short term (Pre-treatment and sorting) Efforts should focus on the development of advanced pre-treatment strategies and robust sorting technologies capable of overcoming the inherent resistance of mixed waste streams. The immediate priority would be to “clean” and standardise the feedstock, rendering it compatible with existing mechanical and thermal processing infrastructure.

4.2.1.2 Medium term (Synergistic integration) Building on these foundations, attention must shift to optimising the interfaces within the hybrid frameworks described in Sect. 3. This includes scaling integrated systems such as chemo-biological and bio-electrochemical platforms. Critically, achieving economic viability at this stage also necessitates upstream product design changes such as eliminating additives that deactivate catalysts or inhibit microbial activity to ensure effective coupling across processes.

4.2.1.3 Long term (Molecular design and integrated ecology) Ultimately, the implementation of Design for Disassembly at the molecular level will enable a step change in circularity. In this paradigm, polymers will be engineered with embedded “trigger” mechanisms (e.g. light- or redox-sensitive linkages) that initiate degradation only under specific industrial conditions. This approach supports the deployment of large-scale, fully integrated recycling ecosystems capable of maximising resource recovery across diverse plastic streams.

4.2.2 Economic implications

Mandating that manufacturers assign products to these specific pathways will foster unprecedented research and development and competitiveness. By streamlining the waste stream, this approach transforms plastic waste into a reliable economic asset, driving capital investment and creating new service

industries focused on efficient, scalable hybrid recycling rather than fragmented waste management.

4.3 Adaptive policy (Standards): from material properties to system compatibility

The establishment of internationally harmonised standards represents the critical regulatory bridge between scientific innovation and global implementation. While foundational frameworks developed by the International Organisation for Standardisation (e.g. ISO 15270, 17,088, 18,604), American Society for Testing and Materials (e.g. ASTM D6400, D7081, D7473), Standards Australia (AS 4736, 5810, 14021), and the European Committee for Standardisation (EN 15343, 15347) provide methodologies for assessing recyclability and biodegradability, they remain fragmented and, in many cases, insufficient.

A fundamental disconnect persists in which many existing standards validate degradation under idealised laboratory conditions that bear little resemblance to the operational realities of industrial waste management infrastructure (Afshar et al. 2024). As a result, certification outcomes often fail to translate into real-world performance, creating space for “greenwashing”. Products labelled as biodegradable may persist in practice because local facilities such as industrial composting or anaerobic digestion systems do not replicate the tightly controlled temperature, residence time, or microbial conditions assumed in the certification protocols.

4.3.1 The strategic shift

To drive a circular economy, standards must move beyond verifying static material properties (e.g., whether a plastic is biodegradable) toward assessing dynamic process compatibility namely, whether a given plastic will reliably degrade within defined industrial systems, such as a 50 °C enzymatic reactor operating over a 48-h timescale. Certification should also ensure process-specific validation by directly linking a product to a defined end-of-life pathway. Rather than applying a generic “compostable” label, standards must verify performance under the residence times and operational conditions of real-world commercial facilities, whether anaerobic

digestion, enzymatic depolymerisation, or chemical solvolysis.

Achieving this shift from material-centric labelling to dynamic process validation requires a structured evolution of the existing standard architectures. To guide this transition, we propose the following tiered research priorities:

Short term: Create product-specific standards that clearly distinguish between inputs for resource recovery (monomer harvesting) versus complete mineralisation (conversion to CO₂ / biomass).

Medium term: Establish process-specific protocols for emerging technologies, such as quantifying yields for enzymatic recycling or pyrolysis oil quality.

Long term: Formulate ecosystem-specific standards to ensure compatibility with both natural environments and future engineered infrastructure.

Finally, harmonisation is essential to prevent “jurisdictional arbitrage,” where low-quality plastics are exported to regions with relaxed testing requirements. Developing rapid and robust compliance testing methodologies such as spectroscopic verification of additives is critical for real-time auditing of waste streams. By aligning product design with verifiable degradation pathways, these mandatory standards will provide the market confidence necessary to transition from linear disposal to circular resource recovery.

4.4 Digital logistics (Infrastructure): the digital and physical backbone

Even with optimisation of the biological toolkit and the deployment of advanced hybrid reactors, the circular economy will fail if waste cannot be efficiently collected, aggregated and sorted. The current dominant model, a centralised hub-and-spoke system reliant on the long-distance transport of low-density, contaminated waste, is both economically fragile and carbon intensive. In many cases, the logistical carbon footprint associated with transporting uncompressed plastic waste exceeds the emissions savings achieved through recycling (Sevigné-Itoiz et al. 2015). Consequently, a fundamental redesign of both the physical infrastructure (logistics networks and processing facilities) and the digital infrastructure

(data tracking and material identification systems) is required to support the hybrid frameworks proposed in this roadmap. This requires a structural transition from centralised mega-plants to a network of smaller, more versatile processing hubs. In contrast to conventional chemical refineries, which necessitate capital-intensive infrastructure to manage high-energy processes, the enzymatic and biological reactors detailed in Sect. 3 function under mild reaction conditions (Sheldon 2016). This thermodynamic advantage permits modular scalability, enabling the deployment of decentralised units closer to waste generation sources. For instance, this proximity allows urban centres to manage specific waste profiles locally. For example, textile-specific enzymatic units can be collocated within fashion manufacturing districts. Functioning as ‘micro-factories’, these distributed units mitigate the carbon penalty of transporting low-density waste, enabling the direct re-integration of monomers back into local manufacturers.

To direct these localised waste streams to their optimal processing technology whether mechanical, chemical, or biological, sorting infrastructure must evolve beyond simple optical recognition. The implementation of Digital Product Passports (DPPs), a central pillar of the European Union’s Ecodesign for Sustainable Products Regulation (ESPR), offers a solution by creating an “Internet of Waste” (Langley et al. 2023). This digital layer is exemplified by the HolyGrail 2.0 initiative, which pilots digital watermarks unnoticeable to the human eye but readable by high-speed sorting cameras (Alastair et al. 2015). By instantly accessing a product’s full composition data, facilities can distinguish between chemically identical but legally distinct items such as separating food-grade PET from non-food cosmetic bottles and automatically route them to the appropriate mechanical or bio-recycling line. For higher-value durable plastics, blockchain integration can further provide an absolute history of ownership and additive content, ensuring that hazardous legacy chemicals are identified before they contaminate sensitive biological systems (Bułkowska et al. 2024).

Finally, this digital and physical transformation must be underpinned by robust economic drivers to ensure that the collection network operates with efficiency comparable to the distribution network. Achieving this requires a reallocation of costs through strengthened Extended Producer Responsibility

(EPR) schemes, in which fees are eco-modulated according to a product's recyclability and explicitly directed toward funding the reverse supply chain (Lifset et al. 2023). The effectiveness of such logistical control is demonstrated by Deposit Return Schemes (DRS), such as Germany's *Pfand* system, which achieves return rates exceeding 98% for PET bottles (Sidorczuk-Pietraszko et al. 2025). Extending this high-capture model to more complex waste streams potentially through the deployment of "smart bins" that read digital product passports and issue consumer credits represents a promising pathway to securing the clean, predictable feedstocks required for high-performance bio-recycling (González et al. 2025).

In summary, the success of this roadmap hinges on the precise alignment of four critical domains. Innovation in isolation whether a novel enzyme or an advanced sorting algorithm is insufficient to close the loop. Instead, the proposed framework demands a tightly coupled system in which material design (Sect. 4.1) directly informs the selection of hybrid processing technologies (Sect. 4.2). This alignment must be rigorously validated through adaptive policy and standards (Sect. 4.3) and operationalised *via* digital logistics (Sect. 4.4). Only by integrating the material, the process, the data, and the policy into a unified framework can the system transition from linear disposal to a robust circular economy, transforming plastic waste from an environmental liability into a predictable, high-value industrial asset.

5 Concluding remarks: toward a synchronised bio-industrial economy

The global plastic crisis is fundamentally one of systemic misalignment. Advances in polymer chemistry, material functionality, and product diversification have far outpaced the evolution of waste management infrastructure, locking society into a linear "produce–use–dispose" paradigm in which material design and end-of-life fate are effectively decoupled. As demonstrated throughout this review, no single intervention, whether mechanical recycling, chemical depolymerisation, or biodegradation can adequately address the chemical heterogeneity, contamination, and dispersal characteristic of post-consumer plastic waste. A durable solution therefore requires a

paradigm shift: from isolated technologies toward an integrated framework that couples molecular-level material design with advanced hybrid bio-industrial processing strategies.

To operationalise this shift, two strategic requirements need to be distinguished: Resource recovery (Circularity) and safe mineralisation (Environmental resilience). The former seeks to reclaim high-value monomers and platform chemicals for the circular economy, while the latter ensures that materials unrecoverable due to leakage or contamination degrade benignly in natural ecosystems. Achieving these two goals requires the precise synchronisation of the four domains outlined in our strategic roadmap: Material design (Categorisation): Existing resin identification and recycling codes, rooted in manufacturing convenience, are insufficient for guiding end-of-life outcomes. They must be replaced by a design-for-degradation paradigm in which polymers are synthesised to align with specific and verifiable degradation pathways, mechanical, abiotic, or biotic, under defined environmental or industrial conditions. Hybrid processing (Technology): The limitations of isolated processing methods necessitate a shift toward hybrid frameworks, integrating the high selectivity of biological systems with the throughput of chemical engineering. Synergistic combinations of enzymatic catalysis with thermal, photochemical, or electrochemical pre-treatments offer a realistic pathway for handling complex, mixed, and contaminated waste streams that currently remain outside the reach of conventional recycling systems. Adaptive policy (Standards): Regulatory frameworks must evolve from certifying static material properties to validating dynamic system compatibility. Harmonised international standards should assess real-world material fate across industrial and environmental contexts, ensuring that claims such as "biodegradable" or "compostable" are tied to demonstrable residence times, degradation kinetics, and ecological outcomes. Digital logistics (Infrastructure): Linear, centralised waste logistics must transition toward decentralised, data-enabled networks. Enabled by digital product passports and molecular tagging, the 'Internet of Materials' creates the necessary conditions for economies of scope, where waste is sorted with molecular precision and valorised locally in decentralised hubs.

5.1 Future outlook

The challenge of implementation is no longer strictly scientific; it is industrial and economic. Recognising that the global petrochemical industry cannot be dismantled overnight, the shift to circularity demands an ‘evolutionary revolution’ that balances immediate retrofits with long-term systemic change. Short-term gains will come from advanced sorting and hybrid retrofitting of existing assets, while long-term resilience depends on “Design for Disassembly” at the molecular level. Success hinges on a collaborative model where regulators and industry co-develop the standards that turn “green design” from a liability into a commercial advantage.

The management of end-of-life plastics is at an inflection point; by integrating the material, the process, the data, and the policy into a unified framework, we can transition from a liability management model to a bio-industrial asset recovery model. In this future state, plastic waste will not be viewed as a pollutant, but as a predictable, chemically rich feedstock that fuels a robust and regenerative circular economy.

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Declarations

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